Ryerson University

Department of Electrical & Computer Engineering
Faculty of Engineering & Architectural Science

Course Title:	**CMOS Mixed-Signal Circuits and Systems**
Course Number:	**EE8507**
Semester/Year:	**W2026**

Instructor:	**Dr. Fei Yuan**

Assignment/Lab Number:	**Design Project #1**
Assignment/Lab Title:	**Clock Generator**

Submission Date:	**Feb 14, 2026**
Due Date:	**Feb 14, 2026**

Student LAST Name	Student FIRST Name	Student Number	Section	Signature*
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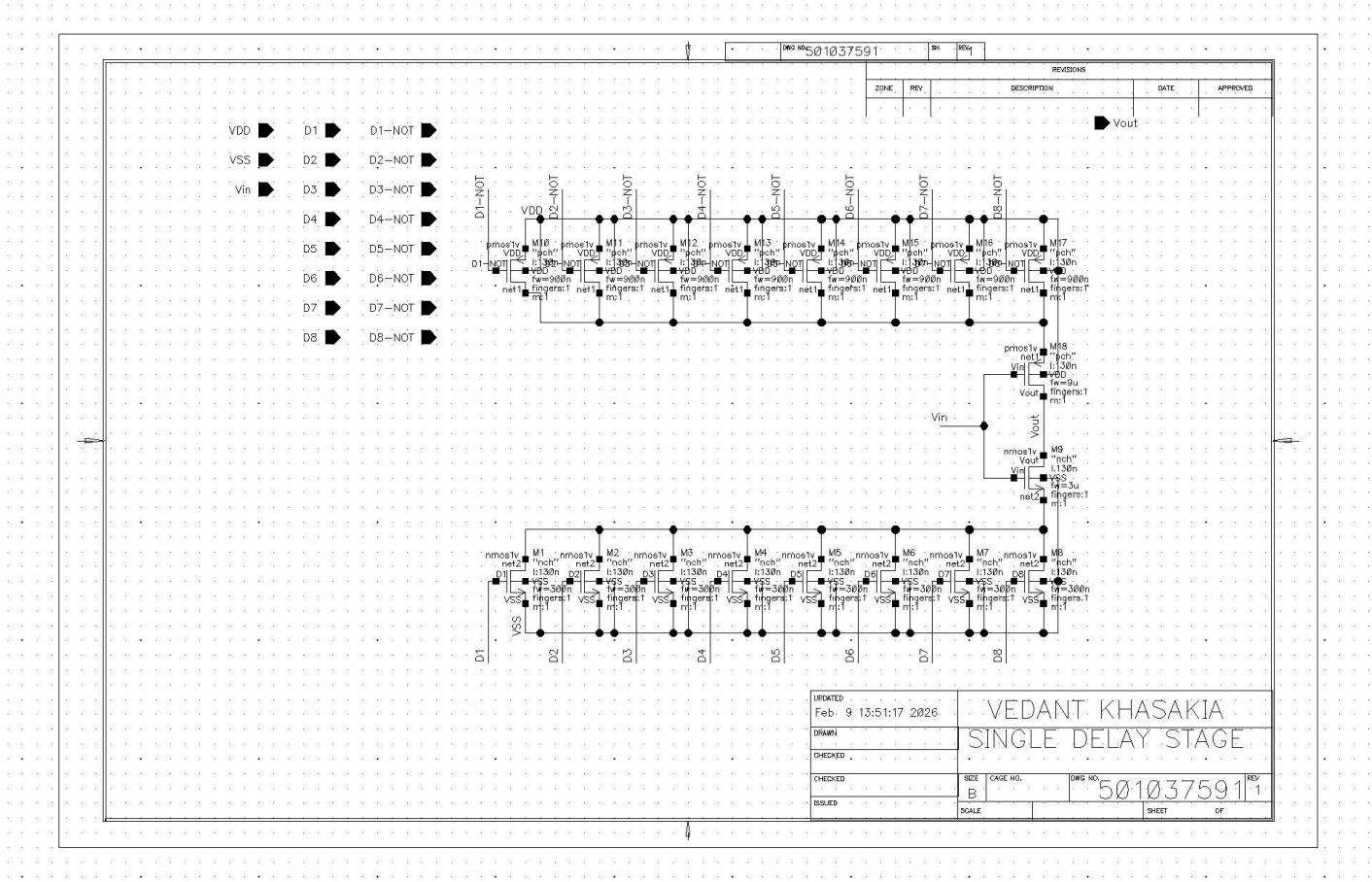
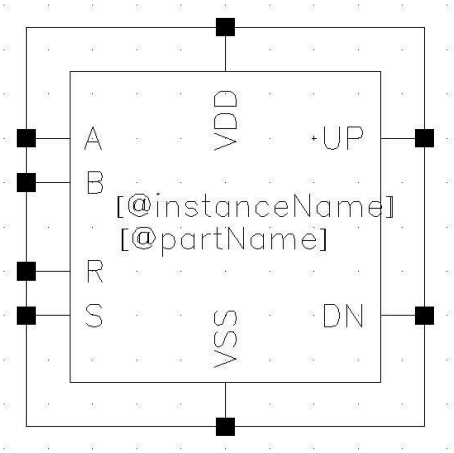
*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

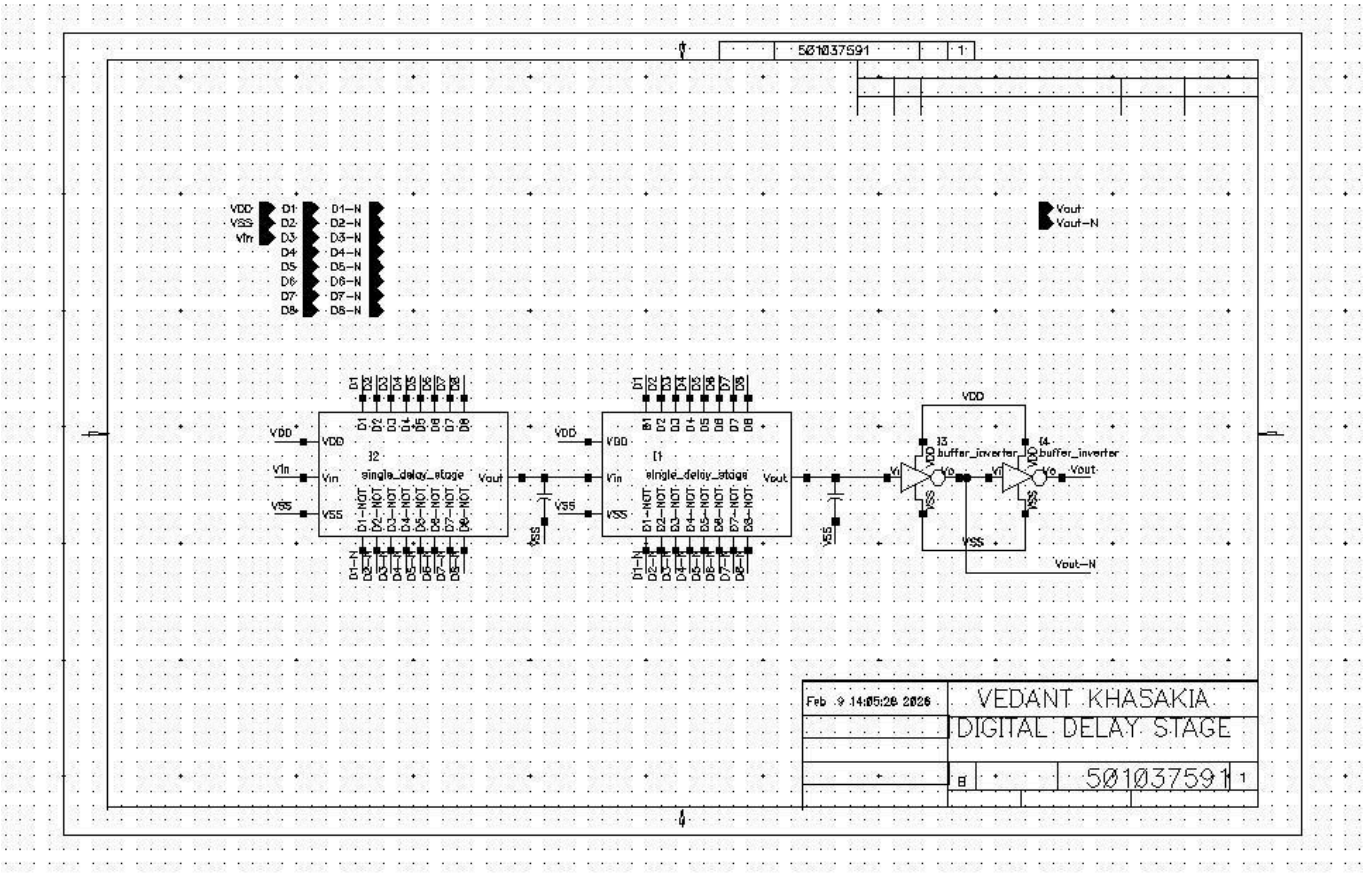
Schematics and Parameters:

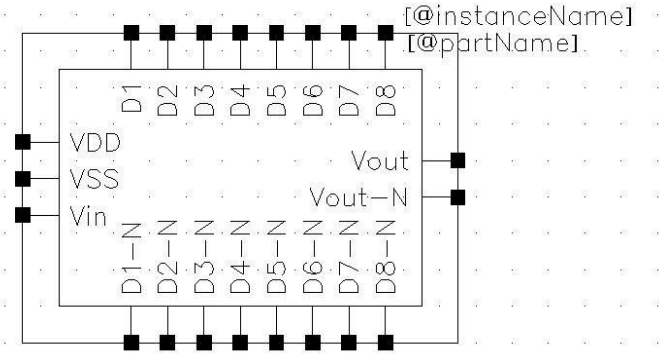
1.1) Single Delay Stage Schematic and Symbol:



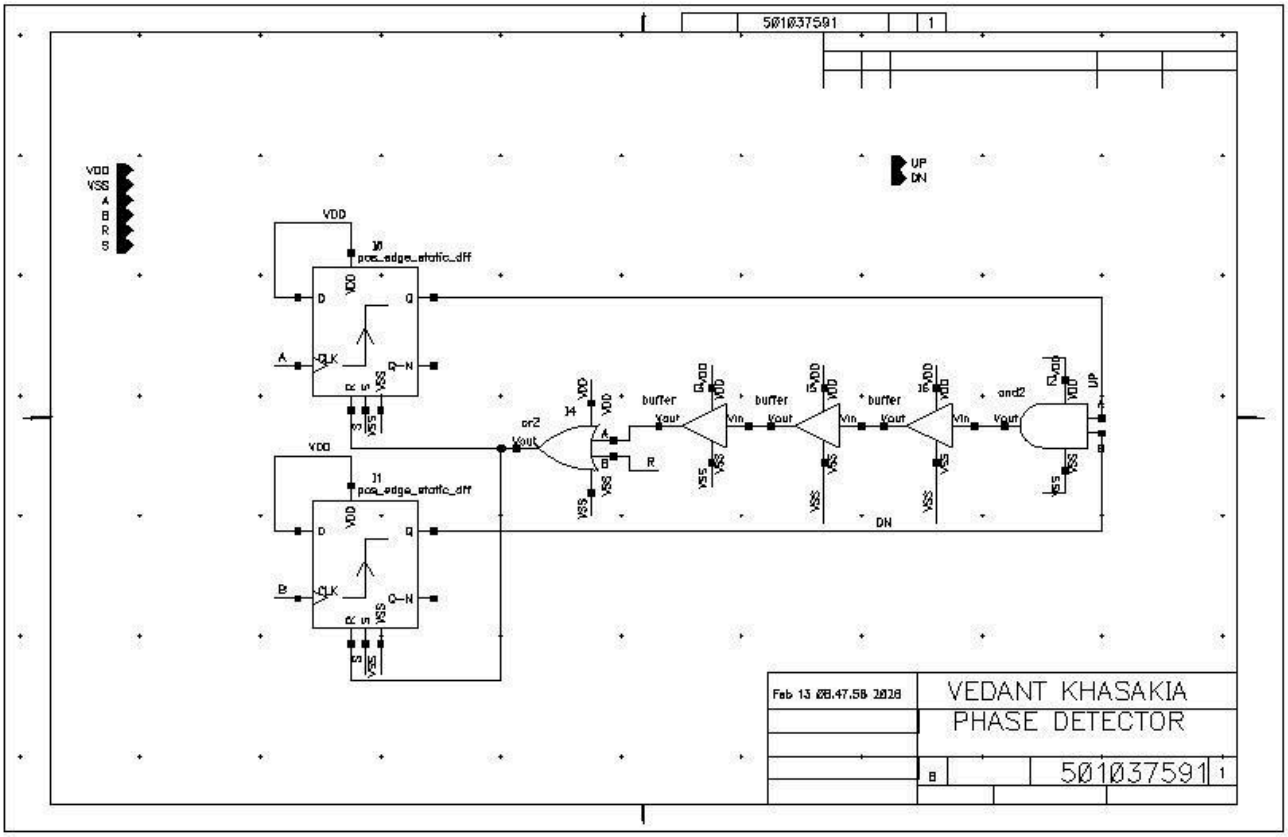


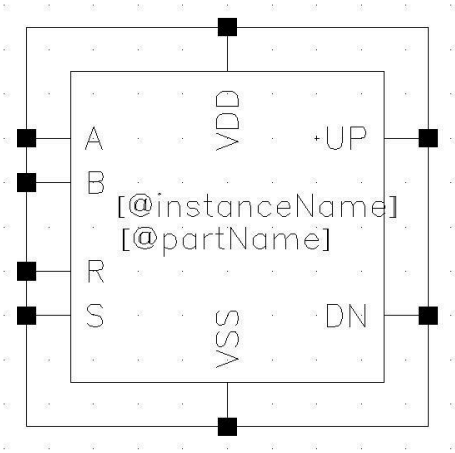
2.1) Single Delay Line Schematic and Symbol:



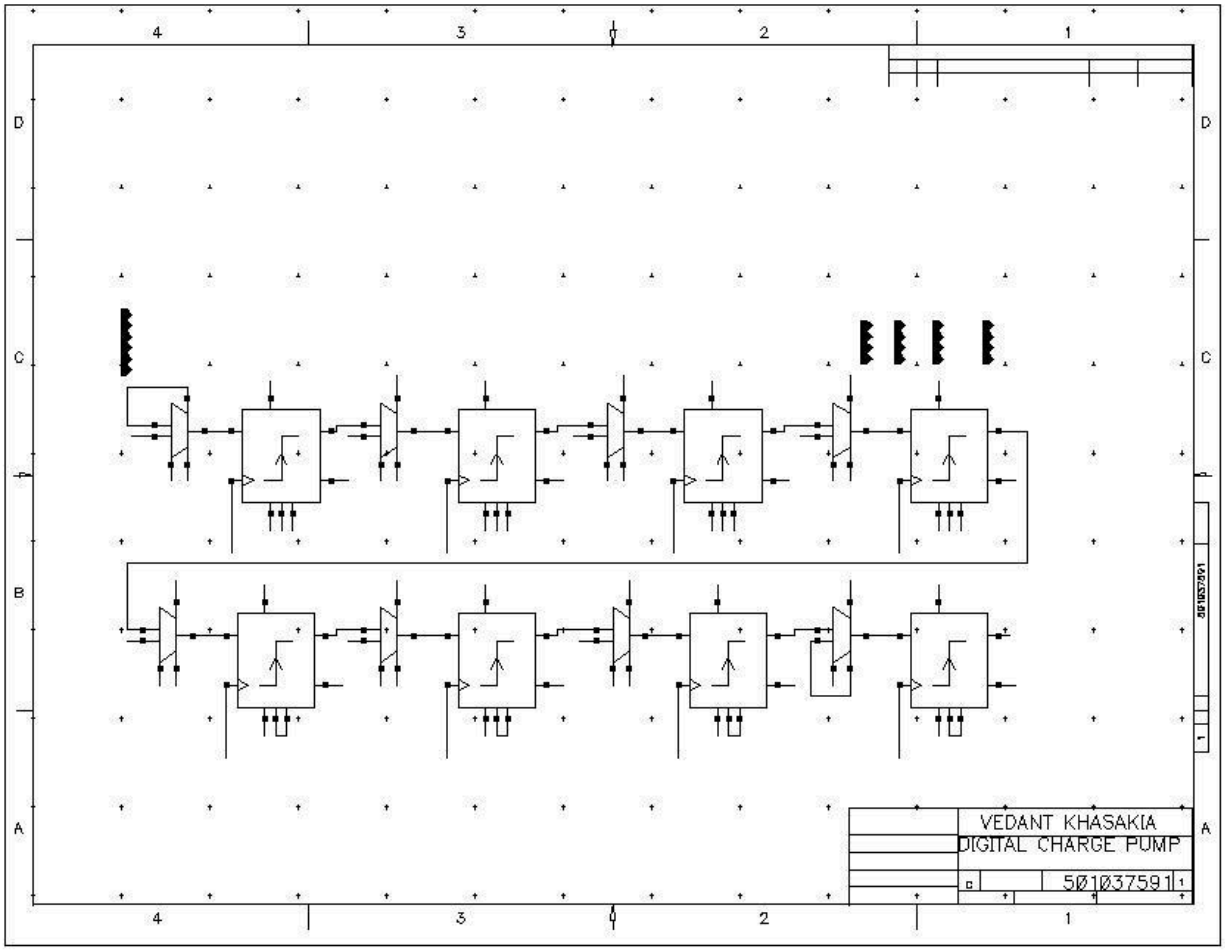


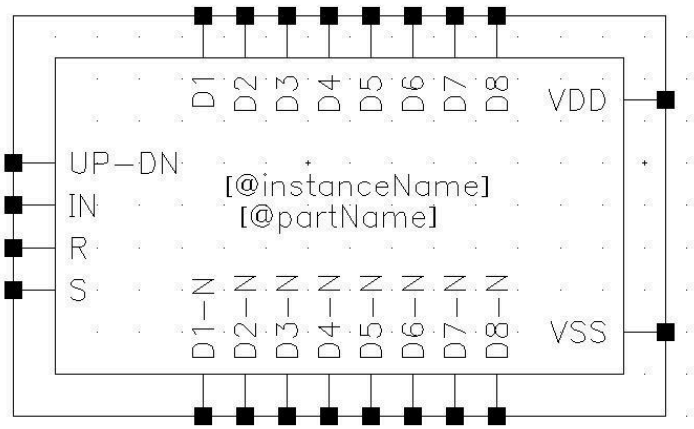
3.1) Phase Detector Schematic and Symbol:



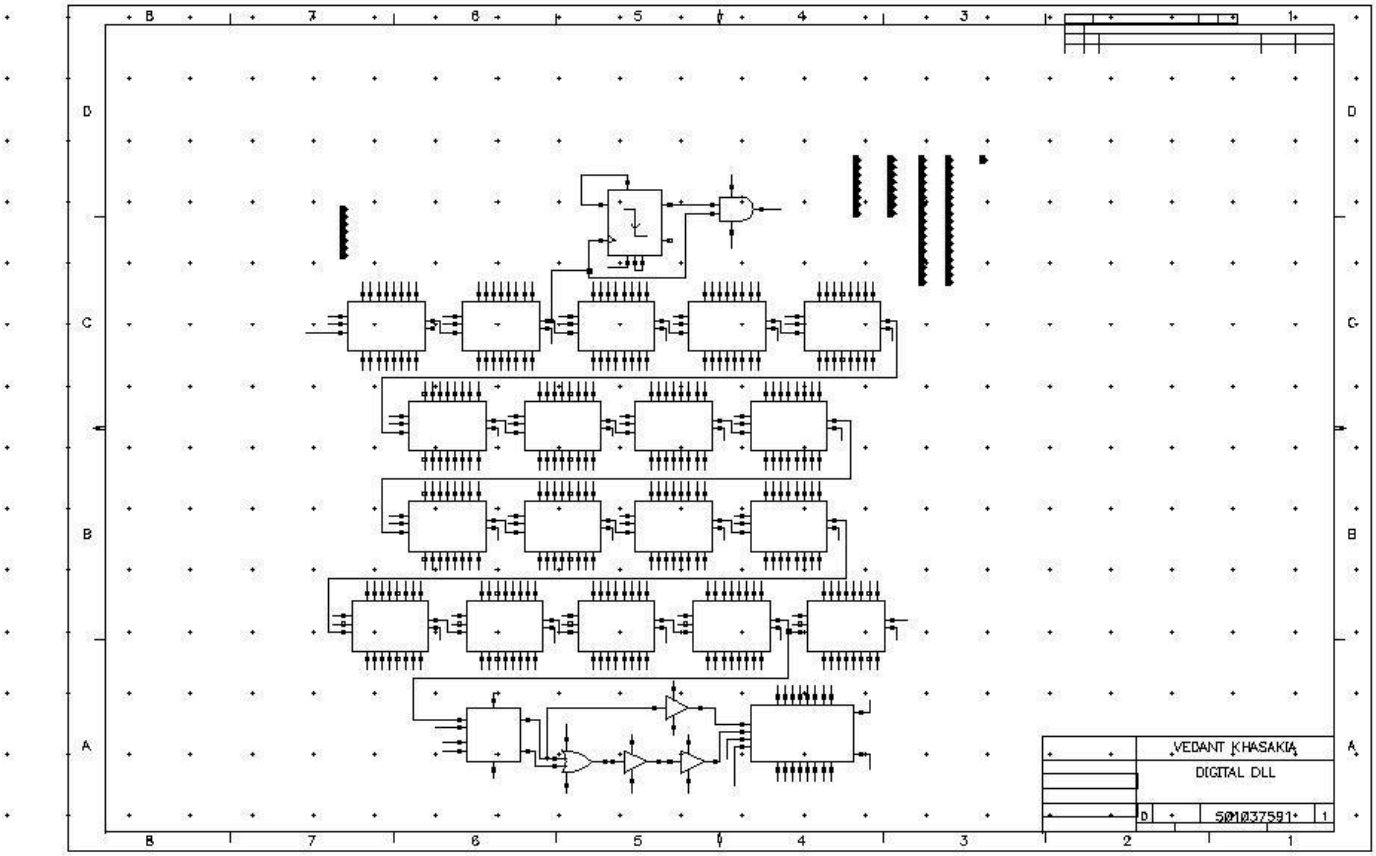


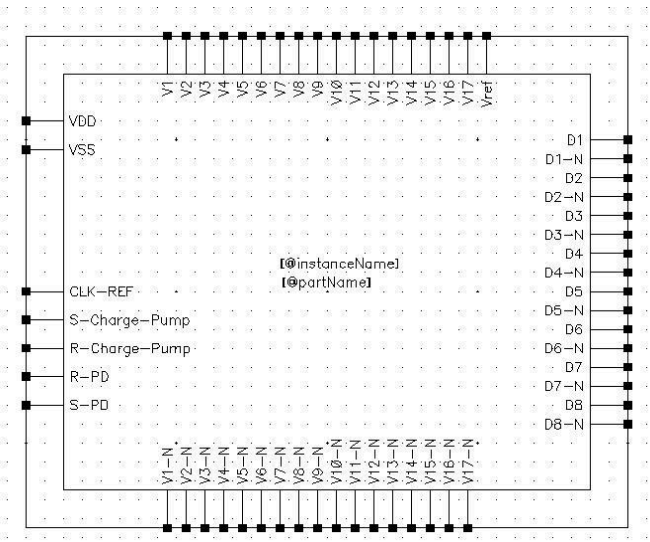
4.1) Digital Charge Pump Schematic:



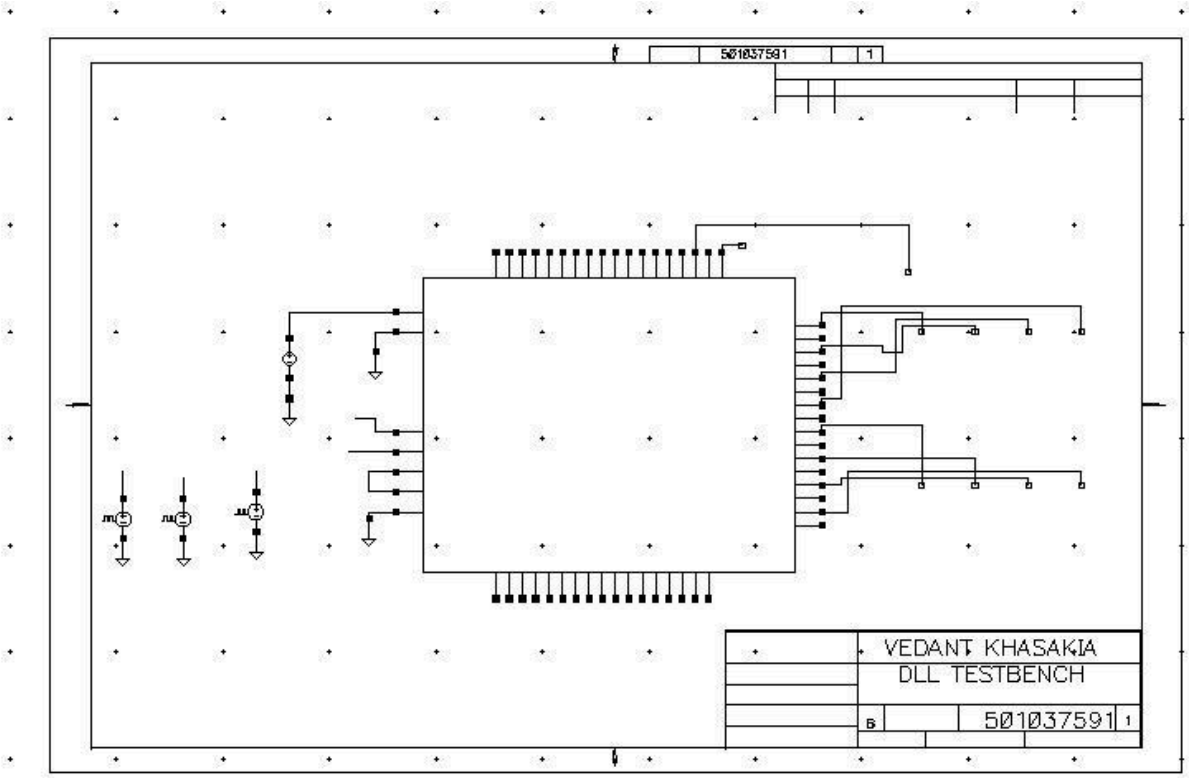


5.1) DLL Schematic and Symbol:



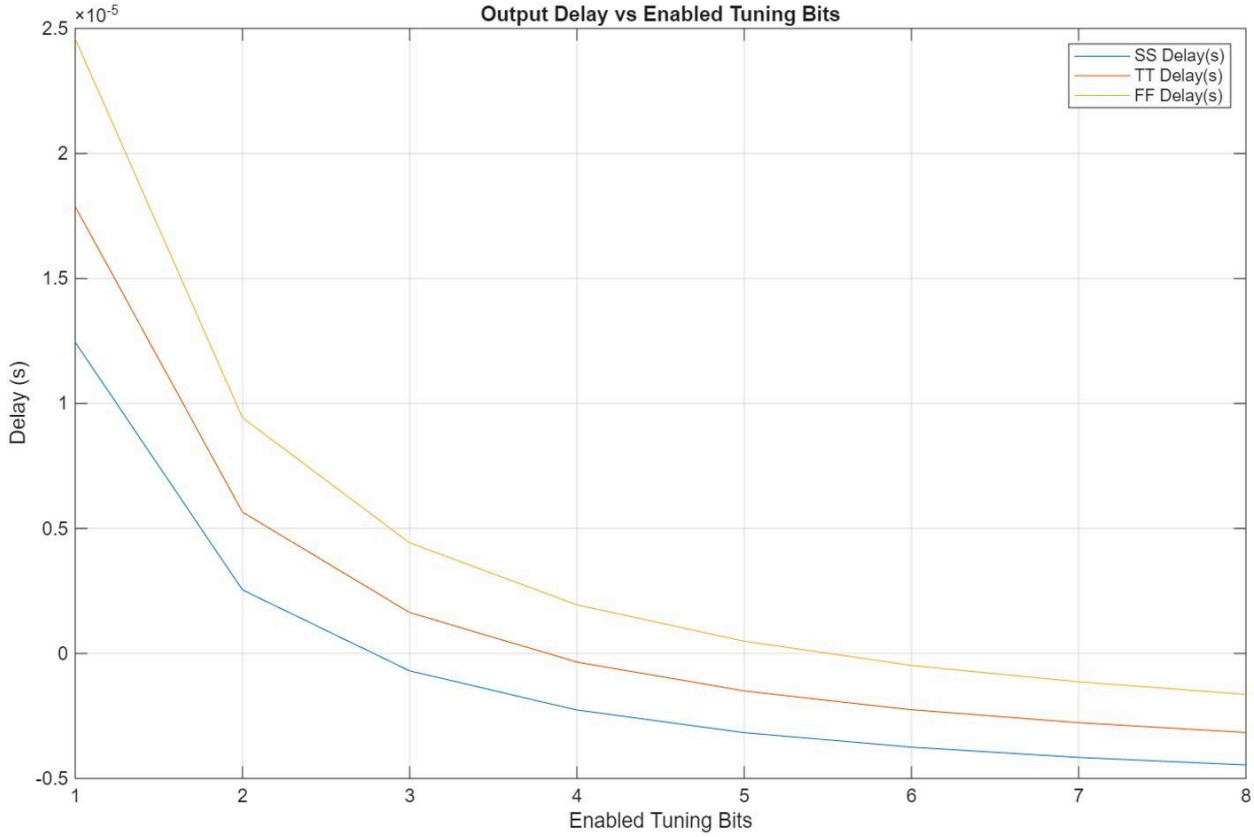


5.2) DLL Testbench Schematic and Symbol:

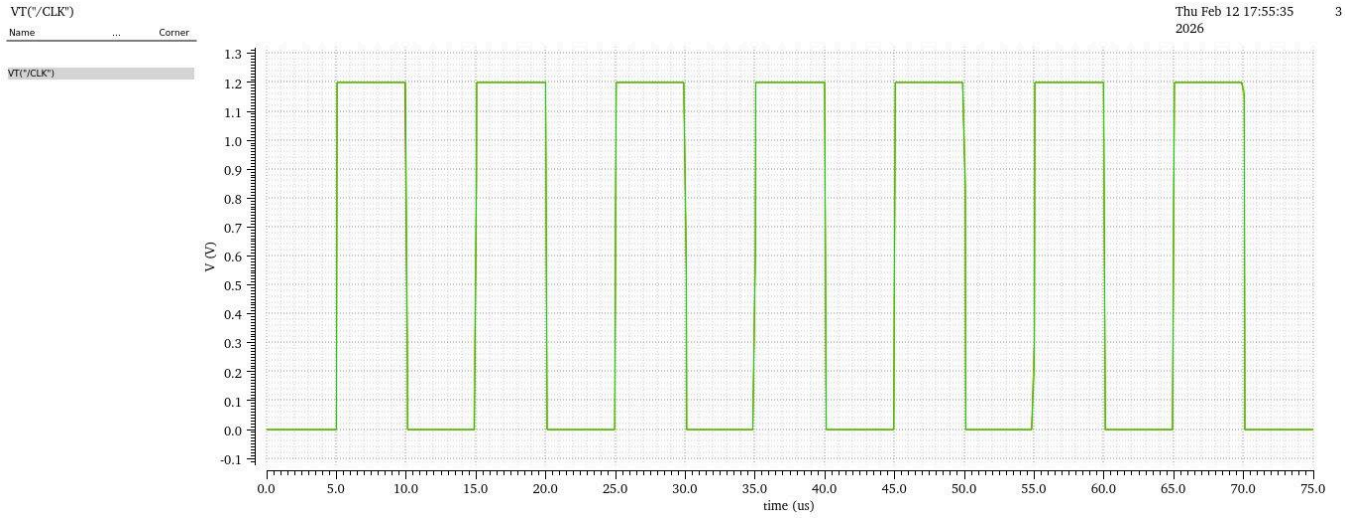


Simulation Results:

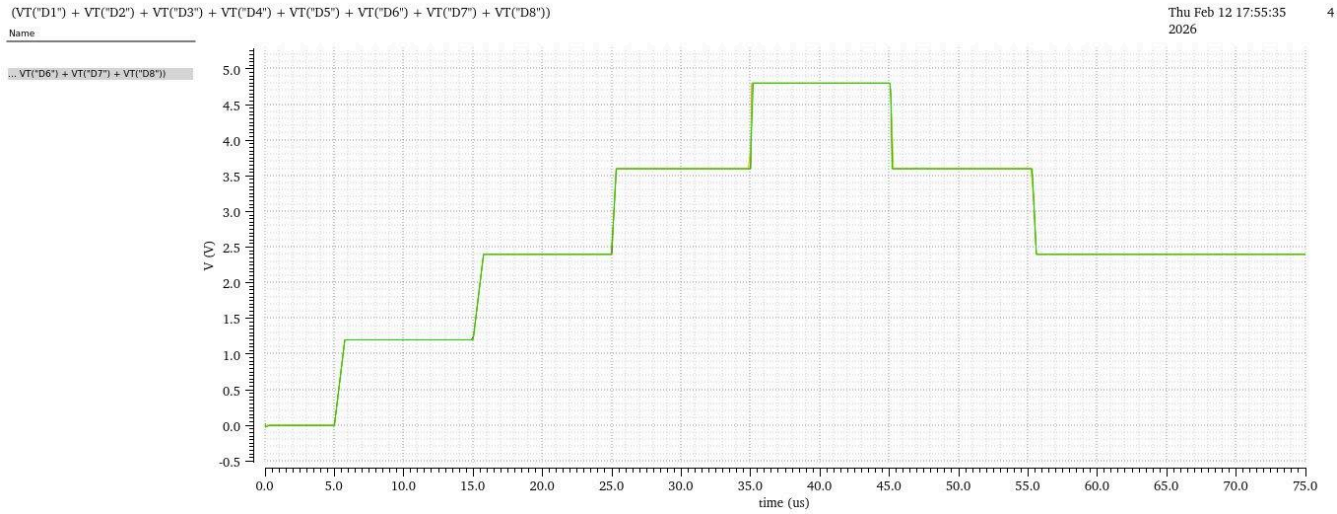
6.1) Delay Stage Test Delay vs Tuning Bit Sum:



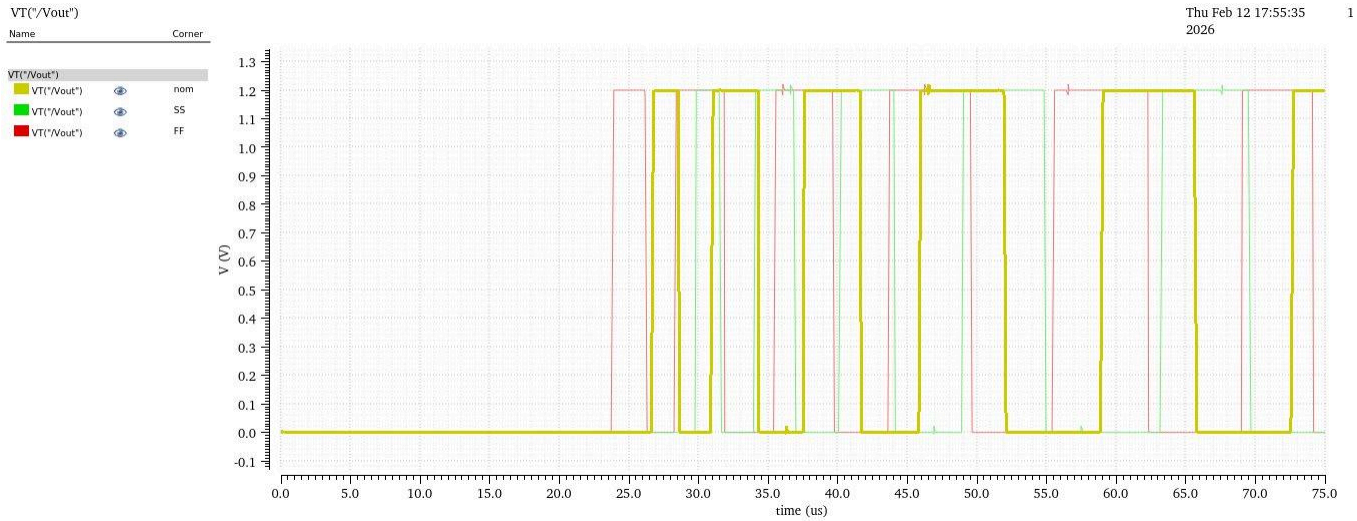
7.1) Digital Charge Pump Test Trigger Signal vs Time:



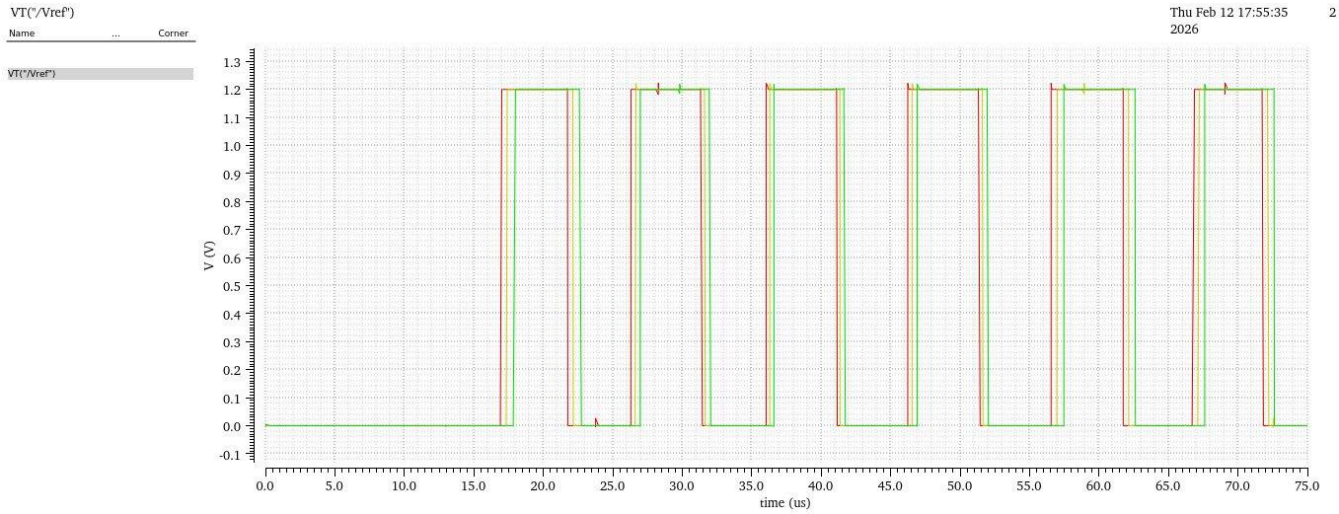
7.2) Digital Charge Pump Test Tuning Bits Sum (UP=3 DN=2) vs Time:



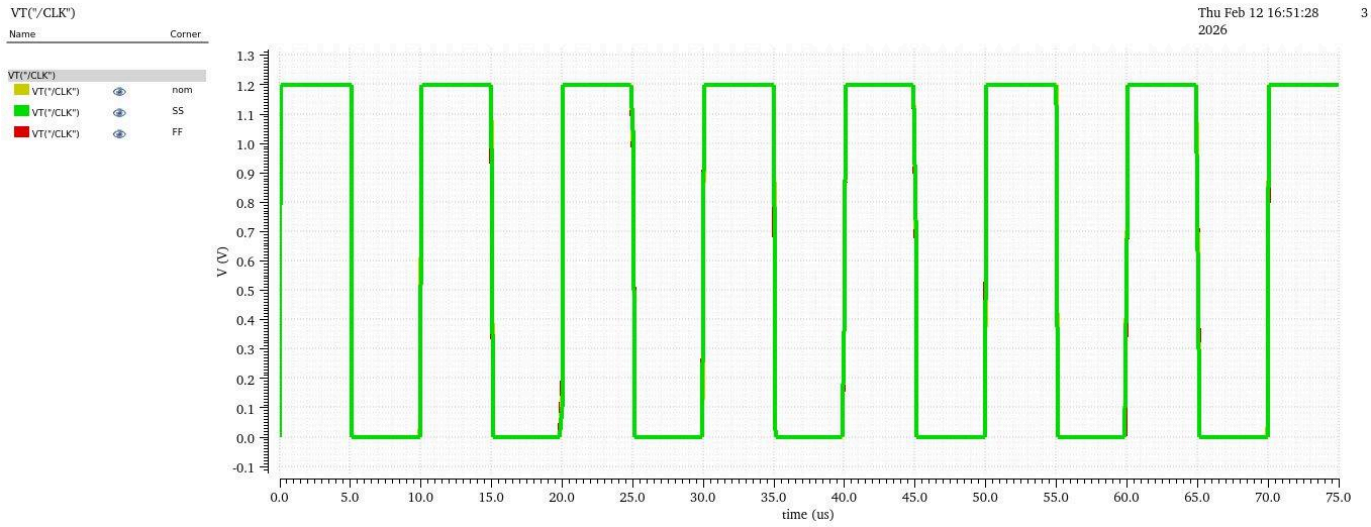
7.3) Digital Charge Pump Test Vout vs Time: (SS = Green, TT = Yellow, FF = Red)



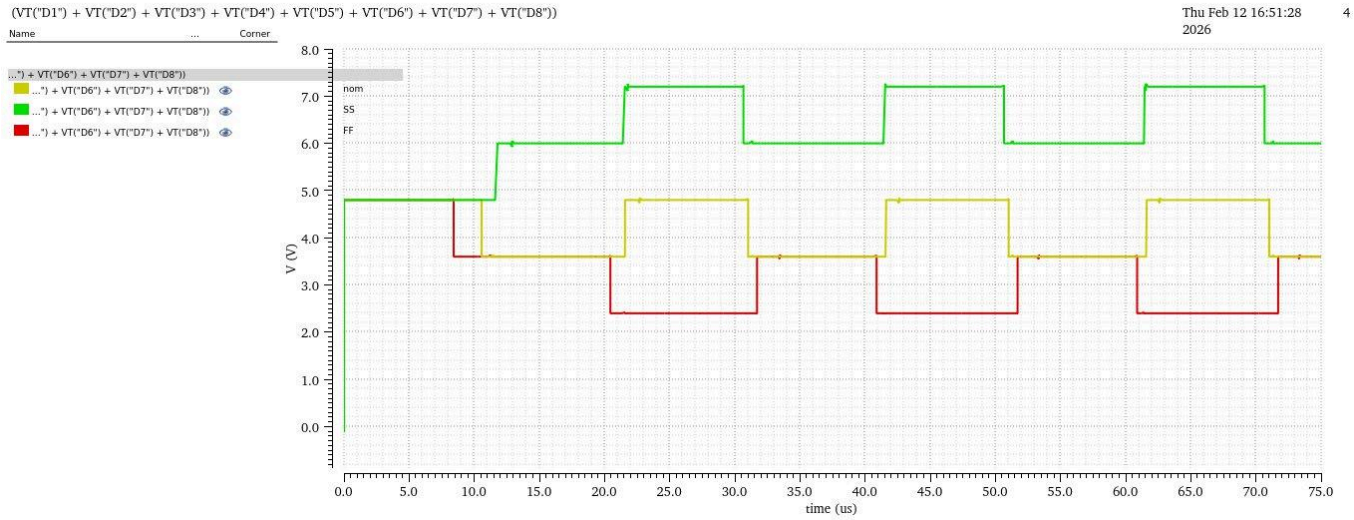
7.4) Digital Charge Pump Test Vref vs Time: (SS = Green, TT = Yellow, FF = Red)



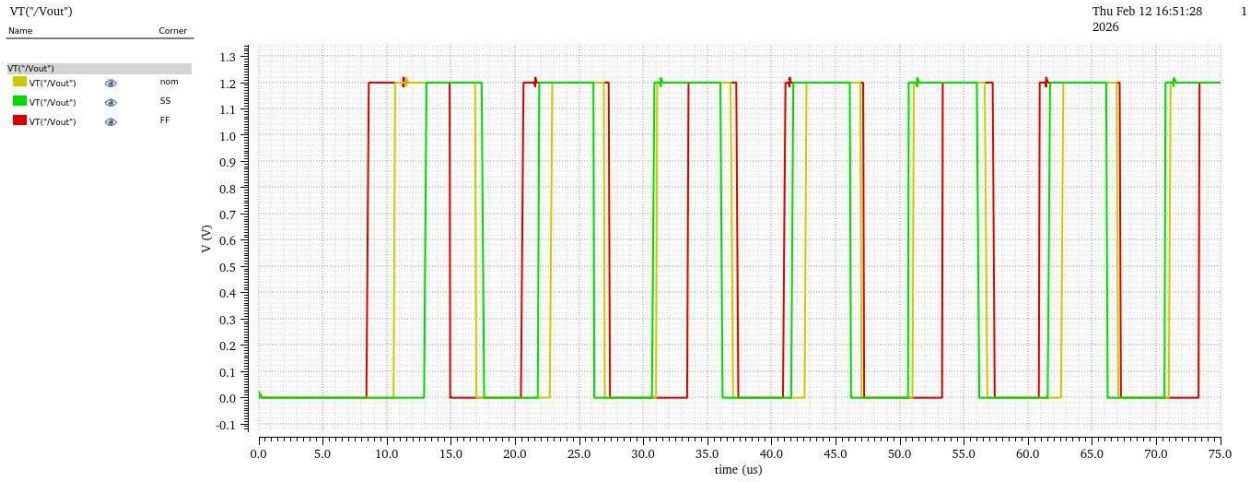
8.1) DLL Test CLK Signal vs Time:



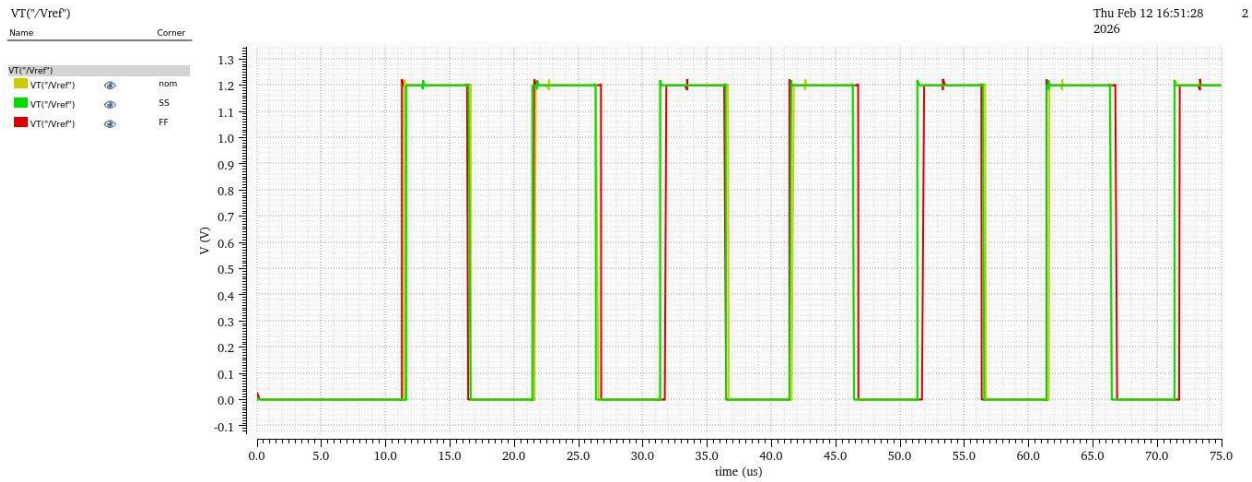
8.2) DLL Test Tuning Bit Sum vs Time: (SS = Green, TT = Yellow, FF = Red)



8.3) DLL Test Vout vs Time: (SS = Green, TT = Yellow, FF = Red)



8.4) DLL Test Vref vs Time: (SS = Green, TT = Yellow, FF = Red)



Conclusions:

This lab successfully demonstrated the critical role that DLLs play in mixed-signal applications and digital signal integrity. This was done by analyzing the different blocks that come together to make a DLL such as the digital delay line, phase detector, and digital charge pump. By slowly enabling the tuning bits (fig. 6.1), a non-linear relation between the delay at the DLL output and the number of enabled tuning bits is depicted, likely due to the capacitive components of the delay line (such as the NMOS caps). A transient analysis of the output with 3 UP and 2 DN signals being fed into the DLL showed how the output changes depending on the number of tuning bits being enabled. Enabling more tuning bits causes the per stage delay of the delay line to decrease, causing the phase to change. A transient sweep of the DLL output with a reference clock being fed into the input demonstrated how the DLL managed to achieve lock and align the output phase with the reference phase. The tuning bits are turned on and off until the lock is achieved causing one tuning bit to oscillate. The different corners (SS, TT, FF) that were used during each analysis demonstrated how process variations and temperatures can affect the outputs of the DLL. This is important as in real life applications, all these variations need to be accounted for.